

The Price of Learning: Experimental Evidence on Agricultural Technology Trials

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Abstract

Free distribution of new agricultural inputs, often through trial packs, is widely used to encourage smallholder farmers to experiment with unfamiliar technologies before deciding whether to adopt them. Yet critics argue that charging even a modest price may improve the effectiveness of such interventions by screening for interested users, increasing commitment through sunk costs, or signaling product quality. This raises a practical question for program design: should trial packs be distributed free of charge, or does charging a positive price make such interventions more effective? We address this question in a field experiment among smallholder farmers in Uganda who were offered trial packs of a recently released maize variety. Our design separately identifies the screening, sunk-cost, and signaling effects of price. We find that willingness to pay predicts subsequent use of the trial pack, suggesting that it can help identify farmers most likely to benefit. Charging a positive price, however, neither increases adoption in the following season nor do we find evidence that it signals product quality. Instead, farmers who received seed for free managed the trial more carefully. Although charging a positive price does not increase adoption, higher willingness to pay predicts higher production in the following season, likely because most farmers recycle seed from their trial harvest. For policy, the value of charging a modest price lies in screening, not in stimulating adoption.

Keywords: Technology adoption; Trial packs; Screening; Sunk cost; Price signaling.

JEL-codes: C93, D83, O12, O13, Q16.

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1 Introduction

Prices are ubiquitous in economic transactions. They are central to the efficient allocation of scarce resources within a society and provide important supply-side incentives. But charging the (full) price may not always be optimal. For instance, if a product or technology is new, providing it for free or at a discount for a short period of time may be necessary to encourage potential users to try it and learn from it (Bawa and Shoemaker, 2004). From a social welfare point of view, subsidies may be justified to ensure access to essential goods and services for poor or disadvantaged communities that may benefit most (Banerjee and Duflo, 2011). Additionally, providing an initial bundle of subsidized goods or services can help overcome barriers to adoption by giving a temporary boost that enables reinvestment and sustained use over time, potentially setting off a path to long-term welfare gains, in a process reminiscent of the dynamics described in the poverty trap literature (Balboni et al., 2021; Visser, Jumare, and Brick, 2020). More broadly, from a policy perspective, in the early stages of market development, public subsidies can play a catalytic role by stimulating initial demand for promising technologies or services, helping them reach a critical mass of users that enables economies of scale and attracts sustained private sector engagement. Finally, the presence of positive externalities provides another strong rationale against charging the full price for socially beneficial goods and services (Miguel and Kremer, 2004).

At the same time, there are concerns that free or subsidized provision of goods and services can diminish their perceived value among recipients. When this occurs, such goods may remain unused, be repurposed, or even resold. For instance, free bed nets have sometimes been used for fishing, and subsidized chlorine for drinking water treatment has occasionally been diverted to household cleaning (Cohen and Dupas, 2010; Ashraf, Berry, and Shapiro, 2010). This tension is especially salient in the design of public health and agricultural interventions, where concerns about cost recovery, wastage, and behavioral responses to subsidies often shape policy. It raises a first-order question for the design of trial-pack and subsidy programs: should a new agricultural technology be given away, or does charging a (modest) price make such interventions more effective, and if so, through which channels?

There are at least three mechanisms through which charging a positive price for a good or service may increase its uptake and intended use. First, prices can act as *signals* of unobserved product attributes, such as quality, effectiveness, or expected benefits. For example, in contexts of uncertainty and asymmetric information, consumers may infer higher quality from higher prices, increasing the extent to which they value and use the product (Milgrom and Roberts, 1986; Bagwell and Riordan, 1991). Second, a *screening* effect may arise: individuals who place a higher intrinsic value on the product are more likely to be willing to pay for it, leading to better targeting of actual users. Third, a *sunk cost* effect refers to a psychological commitment whereby individuals are more likely to value and use a product simply because they have incurred a cost to obtain it, regardless of its objective utility (Kahneman and Tversky, 1979; De Bondt

and Makhija, 1988).

A key challenge is that these mechanisms are typically conflated in observational data, as each mechanism implies a positive correlation between price and use. Building on [Ashraf, Berry, and Shapiro \(2010\)](#) who employ a two-stage pricing design in which commodities are first offered at randomized prices and then subject to a surprise discount to disentangle screening and sunk cost effects, we extend this design by introducing an additional initial stage that allows us to also isolate signaling effects. Specifically, in the first stage, farmers are randomly assigned an offer price that may serve as a signal of the product’s quality. In the second stage, a bargaining game elicits willingness to pay, which allows us to identify screening effects. In the third stage, a surprise discount is introduced, enabling us to measure sunk cost effects. The contribution of each mechanism on a particular outcome can then be estimated through regression analysis: To capture signaling effects, we assess the impact of the initial randomized offer price while controlling for both willingness to pay and whether the farmer paid for the good or service. To identify the screening effect, we focus on the marginal effect of willingness to pay while controlling for the initial randomized offer price and whether the farmer ultimately paid or received the good or service for free. Finally, to test for sunk cost effects, we focus on the indicator for full payment versus free provision, controlling for willingness to pay and the initial offer price.

We apply our experiment to seed trial packs for a recently released hybrid maize variety, distributed to smallholder farmers in eastern Uganda, a setting broadly representative of how new seed is introduced across much of the region. Studying how pricing may affect adoption and use of such agricultural technologies presents a policy-relevant case for several reasons. First, given the novelty of and limited familiarity with a new variety, trial packs may be necessary to encourage experimentation among risk-averse farmers ([Blackie and Mann, 2005](#); [Mulungu et al., 2026](#); [Foster and Rosenzweig, 1995](#); [Brick and Visser, 2015](#)). Second, social learning, especially through peer networks, plays a critical role in the large-scale adoption of new agricultural technologies, suggesting the existence of significant positive externalities ([Conley and Udry, 2010](#); [Bandiera and Rasul, 2006](#); [Van Campenhout, 2021](#)). Third, seed purchases involve non-trivial upfront costs, making them susceptible to sunk cost effects ([Soman, 2001](#)). Finally, seed markets are often plagued by significant information asymmetries regarding product quality, which adds an additional layer of risk to farmer investment decisions and gives rise to potential price signaling ([Bold et al., 2017](#); [Ahmad, 2022](#)).

The experiment was implemented in eastern Uganda with about 760 smallholder farmers. To assess how pricing affects outcomes along the adoption pathway, we collected detailed data in two rounds: in the short run (the first season after distribution) we tracked whether the seed trial pack was planted, whether seed and harvest were kept separate, who in the household used it, and whether complementary inputs such as fertilizer and chemicals were applied. In the longer run (the subsequent season) we measured adoption of improved and promoted varieties, complementary input use, and production on a ran-

domly selected plot. This dual focus enables us to go beyond the immediate use of the trial pack and examine whether pricing effects persist into subsequent agricultural decisions.

We find no evidence that the initial offer price served as a signal for unobserved product attributes. Willingness to pay, in contrast, is positively associated with more appropriate use of the trial seed, greater investment in complementary inputs, and stronger intentions to adopt improved varieties (including the promoted seed), suggesting that prices may screen for farmers who are better positioned to benefit from new technology. However, paying for the seed *reduces* rather than reinforces appropriate use, contrary to the sunk cost hypothesis: farmers who received the seed for free were more likely to keep the harvest separate, a behavior that facilitates learning and evaluation. While adoption rates in the subsequent season do not differ across treatment arms, screening effects on production persist: since most farmers recycle seed from their trial harvest, pricing-induced differences in first-season management appear to affect the quality of recycled seed and thus subsequent production. Taken together, these results reveal a tension in pricing: while willingness to pay identifies farmers who use the trial seed more appropriately, requiring payment reduces the quality of experimentation. This suggests that eliciting willingness to pay may be valuable as a targeting tool, even if the seed is ultimately provided at a discount or for free.

Our study contributes to a growing literature that examines how pricing affects the uptake and adoption of goods and services. Seminal studies by [Ashraf, Berry, and Shapiro \(2010\)](#) and [Cohen and Dupas \(2010\)](#) offer contrasting insights. [Ashraf, Berry, and Shapiro \(2010\)](#), in a two-stage pricing experiment of chlorine for drinking water purification in Zambia, find evidence of screening effects (suggesting that willingness to pay reflects private information about future use) but no support for sunk cost effects. Their results imply that pricing may improve targeting efficiency, potentially justifying lower subsidy levels. In contrast, [Cohen and Dupas \(2010\)](#) study demand and usage of insecticide-treated bed nets in Kenya and find that even small prices substantially reduce uptake, with little evidence that pricing improves targeting through screening mechanisms. They conclude that free distribution may yield greater health benefits by expanding coverage. More recently, [Mahmoud \(2024\)](#) uses a two-stage pricing experiment for improved wheat variety seed in Bangladesh to examine whether prices can screen farmers based on their returns to the technology. Farmers first face randomized subsidy levels, after which non-buyers are randomly offered the seed for free. This design allows a comparison between self-selected non-buyers and the broader population. The study finds that farmers who initially decline to purchase the seed adopt it when offered for free and achieve realized returns similar to those of the average farmer. These results suggest that willingness to pay does not reliably screen farmers based on returns and that prices may primarily restrict access rather than target the technology to higher-return users.

Our study also touches on a first-order debate in food policy: how to target agricultural input subsidies. Under the market-smart subsidy framework

that has guided a generation of programs in sub-Saharan Africa, a subsidy should reach farmers who would use the input profitably but would not otherwise purchase it, without displacing commercial sales and with a credible exit path (Jayne et al., 2018; Holden, 2019). In practice, the administrative mechanisms used to identify such farmers, community-based targeting and proxy-means tests, have proven costly, regressive, and prone to elite capture (Basurto, Dupas, and Robinson, 2020; Jayne et al., 2018). A prominent reform proposal responds by using price itself as the targeting device: setting a positive farmer contribution so that farmers who value the input above that price, and who tend to have higher returns, self-select into the program, while others take up a cash alternative (Duchoslav and De Weerd, 2022; Banda et al., 2022). This is precisely the screening argument, and it is the same logic we test at the level of an individual technology. Yet the case for price-based self-targeting rests largely on stated willingness to pay, and it sits in tension with the learning rationale that motivates free distribution in the first place, since a positive price that improves targeting may simultaneously dampen the experimentation and careful use that trial packs are meant to encourage (Omotilewa, Ricker-Gilbert, and Ainembabazi, 2019). Our design speaks to both margins with revealed preferences: whether willingness to pay screens for farmers who use the technology well, and whether requiring payment strengthens or weakens that use.

Our study builds on this body of work but adds an important innovation by explicitly isolating and testing signaling effects as a third mechanism alongside the more commonly studied screening and sunk cost effects. Existing two-stage pricing designs typically examine whether prices screen for users with higher private valuations or induce sunk cost-driven commitment, typically by estimating one effect while controlling for the other. However, if prices also signal product attributes in settings characterized by experience goods or information asymmetries, then ignoring this channel may confound estimates of both screening and sunk cost effects.¹ To address this limitation, we extend the canonical two-stage design into a three-part pricing experiment that separately identifies signaling, screening, and sunk-cost effects by randomizing the initial offer price (signaling), eliciting willingness to pay through a bargaining game (screening), and assigning a surprise discount to a randomly selected subsample of participants (sunk-cost effects).

The remainder of this article is organized as follows. We first explain the methods used and the experimental design, followed by the estimation strategy. We then turn to sampling and provide descriptive statistics of our study population. The analysis consists of two parts. We first look at the price elasticity of demand for the seed trial packs using willingness-to-pay data. We then pro-

¹The reason is that signaling shifts willingness to pay; if this is not explicitly controlled for, willingness to pay absorbs part of the signaling effect and becomes endogenous. As a result, the coefficient on willingness to pay (interpreted as screening) will be biased and, because sunk-cost estimates are identified from payment outcomes conditional on willingness to pay, bias carries over to the sunk-cost effect estimate. Both Ashraf, Berry, and Shapiro (2010) and Cohen and Dupas (2010) discuss signaling as a theoretical possibility, but their design primarily focuses on disentangling screening from sunk cost effects.

vide estimates for screening, sunk cost, and signaling effects. The last section concludes and reflects on implications for policy.

2 Methods and experimental design

The standard approach to disentangle screening and sunk cost effects arising from charging a price is the *two-stage pricing design* pioneered by [Ashraf, Berry, and Shapiro \(2010\)](#) and [Cohen and Dupas \(2010\)](#). In a two-stage pricing design, the first stage involves randomizing the posted price of a product across participants or sites, generating variation in who initially decides to purchase. This allows researchers to test for screening effects, since willingness to buy at a given posted price may correlate with expected use. In the second stage, among those who agreed to purchase at the posted price, a subset is randomly offered an unanticipated discount or rebate that lowers the actual transaction price. Because all of these individuals revealed a willingness to pay the original posted price, any systematic difference in subsequent usage between those who ultimately pay more versus less can be attributed to sunk cost effects, rather than to differences in underlying demand. Together, this two-stage design cleanly separates selection at the offer stage from psychological responses to the act of paying.

We build on the standard two-stage pricing design by introducing an additional stage at the beginning, yielding a *three-stage pricing design*. In this additional first stage, farmers are presented with a randomized initial offer price. When a technology is new and its performance uncertain, such prices may be interpreted as signals about otherwise unobserved product attributes, such as quality, effectiveness, or expected benefits. Standard two-stage designs implicitly assume that price variation affects behavior only through screening and sunk-cost mechanisms. By randomizing the initial offer price prior to eliciting willingness to pay, our design allows us to test whether prices also influence behavior through signaling effects that shape farmers' beliefs about the technology. The second and third stages then follow the conventional design: we elicit willingness to pay (WTP), providing the basis for identifying screening effects, and finally, we grant a full (100%) surprise discount to a random subset of farmers, allowing us to test for sunk cost effects.

The experiment is implemented as a scripted bilateral bargaining game involving concessional offers. Farmers are offered the opportunity to purchase a bag of seed from a trained enumerator who follows a standardized bargaining protocol designed to mimic the types of negotiations commonly observed in real-world seed transactions.² The enumerator follows a standard script that was

²In two-stage pricing designs, such as those used by [Ashraf, Berry, and Shapiro \(2010\)](#) and [Cohen and Dupas \(2010\)](#), in the first stage, participants are typically offered a good at randomly assigned price points. A common drawback of this approach is that a substantial share of participants may opt not to purchase the good, which, absent over-subscription, can significantly reduce the effective sample size. Since willingness to pay (WTP) is central to analyzing two-stage pricing designs, alternative elicitation methods like the Becker-DeGroot-Marschak (BDM) mechanism can be employed to mitigate this problem to some extent. In

implemented in Open Data Kit (ODK) on Android tablets.

As part of our first stage, an initial ask price was randomly drawn from a uniform distribution, and presented to the farmer as the price of one bag of seed for a new improved variety.³ The enumerator explains what kind of variety it is and what the advantages are.

We then elicit willingness to pay (the second stage). The farmer has the option to accept the bag of seed at this initial offer price or not.⁴ If the farmer does not accept the initial offer price, the farmer enters into bargaining where they are encouraged to name their first counter bid price. A computer algorithm then determines a counter-offer that the enumerator asks in a second round of negotiation. This new ask price is determined as the farmer's bid price plus 80 percent of the difference between the (initial) ask price and the farmer's bid price (appropriately rounded). This updated (lower) ask price is then compared to the bid price of the farmer. If the price difference between the bid and (updated) ask price is smaller than a (crop specific) threshold, the ODK script instructs the enumerator to sell at the price the farmer bids. If the difference between the updated ask price and the bid price is larger than the threshold, the updated ask price is presented to the farmer, and the farmer gets a second opportunity to accept or reject. If the farmer does not accept, they are encouraged to make a second bid and a third ask price is determined as the farmer's last bid price plus 80 percent of the difference between the last ask price and the farmer's last bid price. Bargaining continues until the farmer accepts an ask price, or the price difference between the last bid and last ask price is smaller than a (crop specific) threshold.⁵

We chose bargaining over the more standard BDM mechanism for a practical reason: in piloting, farmers were confused by the one-shot, take-it-or-leave-it logic of BDM, whereas negotiation is how seed is customarily transacted in this setting, so a bargaining protocol elicits willingness to pay in a way that mirrors farmers' real purchasing behavior.⁶ We made this change during piloting, before any study participant was enrolled, and updated the pre-analysis plan accordingly in February 2023; the original plan had specified a BDM elicit-

its basic form, BDM asks participants to state a bid, which is then compared to a randomly drawn price. If the bid is lower, participants do not receive the product; if it meets or exceeds the price, participants purchase the product at the drawn price. A key advantage of this method is that it inherently includes a surprise discount for those whose WTP exceeds the random price—mimicking the second stage in two-stage pricing designs—see appendix E in [Berry, Fischer, and Guiteras \(2020\)](#).

³For maize seed packs (1kg) in Uganda, initial prices ranged from 9,000 to 12,000 Uganda shillings (UGX) in increments of UGX 1,000.

⁴In our Uganda sample, the negotiated price equals the initial offer price for 118 out of 758 farmers in the seed arm (15.6 percent); 111 of these accepted the initial offer outright, while the remaining 7 returned to it after bargaining. We discuss the implications of this right-censoring, and the corresponding robustness check, below.

⁵To make the bargaining also incentive compatible for the enumerators, we told them in advance that the money that is collected from farmers during this first stage will be distributed equally among all the enumerators.

⁶This design change led us to update the [pre-analysis plan](#), which is reflected in [commit dated Feb 11, 2023: "changed PAP after field testing - dropped BDM for bargaining experiment"](#) on GitHub.

tion. This raises a natural question, however: does the bargaining game elicit true willingness to pay in the same way a BDM mechanism would? In a BDM, truthful revelation is a dominant strategy: participants have no incentive to misreport their WTP regardless of the randomly drawn price. In bargaining, by contrast, the farmer will start below their true WTP in early rounds, in an effort to strategically shade their WTP (Hortaçsu, Kastl, and Zhang, 2018). However, the multi-round alternating-offer structure of our protocol resolves this concern. The final agreed price (whether the farmer accepts the last ask or the algorithm accepts the farmer’s last bid) reflects the point at which the farmer was no longer willing to go higher, and therefore approximates the farmer’s maximum WTP. To the extent that the negotiated price falls short of true WTP because some farmers capture surplus by stopping short of their maximum, this introduces classical measurement error that attenuates our estimates toward zero but does not bias the identification of screening, signaling, or sunk-cost channels.

Because farmers bid with their own money, the negotiated price reflects not only how much a farmer values the seed but also their ability to pay. We consider this a feature of the design rather than a limitation. What a program can elicit in the field is effective willingness to pay: demand expressed under real-world budget constraints. Wealth, liquidity, and complementary resources raise both what a farmer is able to bid and what the farmer stands to gain from adoption, and are therefore part of what a screening mechanism is meant to capture. Eliciting valuations that are purged of budget constraints, for instance by providing participation fees, would introduce hypothetical bias of its own. Paying out of pocket is, moreover, not specific to bargaining: under a BDM mechanism, farmers would equally pay with their own money. In the analysis, we complement the main estimates with specifications that control for baseline proxies of wealth and liquidity (Appendix E). These controls do not purge the estimate of a confound; rather, they indicate how much of the screening effect operates through observable resources and how much through the additional information contained in the negotiated price.

A final concern is right-censoring. For the 118 farmers whose negotiated price equals the randomly assigned initial offer, willingness to pay is only known to be at least as high as that offer, and the screening and signaling channels are collinear. We retain these farmers in the main analysis with their willingness to pay set to the offer price, which attenuates the screening coefficient toward zero and thus makes our screening estimates conservative. As a robustness check, we re-estimate all main specifications excluding them (Appendix C). If anything, the screening effect on trial pack use strengthens when the sample is restricted to farmers whose willingness to pay is point-identified (0.023 against 0.017 in the full sample, $p=0.006$), while the sunk-cost effects on the use index (-0.094, $p=0.028$) and on keeping the harvest separate (-0.125, $p=0.010$) and the screening effect on subsequent production (0.079, $p=0.008$) are essentially unchanged.

After bargaining was concluded, we asked some additional control questions and, near the end of the interaction, offered a randomly selected subgroup of farmers a surprise discount (the third stage). Unlike most two-stage pricing

designs that introduce random or tiered discounts (often to determine optimal subsidy levels), we employ a single, full (100%) discount. This decision is guided by four considerations. First, we anticipate a behavioral discontinuity between paying any positive amount and receiving the good for free, consistent with findings in behavioral economics that zero prices can have a disproportionate impact on uptake and use (eg. [Shampanier, Mazar, and Ariely, 2007](#); [Shukla, Pullabhotla, and Baylis, 2022](#)). Second, concentrating the sample in two distinct groups—those who pay and those who receive the good for free—maximizes statistical power for estimating the behavioral difference between paying and receiving the good, relative to designs that spread observations across many price points. Third, by replicating how new agricultural technologies are often introduced in practice, with free sample packs provided to encourage trial and learning, our approach strengthens the external validity of the study.⁷ Fourth, our design focuses on a single, full discount rather than multiple rebate tiers, ensuring that all variation in whether farmers paid or not is generated directly by randomized assignment. This design isolates the sunk-cost effect at the extensive margin (pay versus free) in a transparent way. By contrast, canonical two-stage pricing designs identify sunk-cost effects from variation in the prices paid among those who choose to purchase. Because purchase itself is endogenous, separating sunk-cost effects from screening requires additional assumptions. Our design avoids this issue by generating exogenous variation directly at the pay-versus-free margin.

Our experimental design generates three sources of variation that allow us to identify distinct price mechanisms. First, the randomized initial offer price may act as a signal of otherwise unobserved product attributes, such as usefulness or quality. Second, the willingness-to-pay (WTP) price, defined as the final price agreed upon during the bargaining process, captures farmers’ willingness to pay and thus provides the basis for screening. Third, we use an indicator of whether the farmer paid the agreed price or received the pack for free to identify the sunk-cost effect.⁸

Seed trial packs used in this study contain 1 kilogram of improved maize variety seed. We use a recently introduced hybrid variety popularly known as Bazooka, produced by Naseco Seed Ltd. The seed is high yielding. It promises between 3.5 and 4 metric tons per acre and was partly chosen because it is widely available on the market. The 1 kg bag of maize seed is enough to plant about one-eighth of an acre.

At the time of the study, a 1 kilogram bag of Bazooka retailed for about UGX 12,000 at agro-input dealers. The randomized initial offer prices in the bargaining game, which ranged from UGX 9,000 to UGX 12,000 (see below),

⁷In [Ashraf, Berry, and Shapiro \(2010\)](#), the possibility of non-linearities around a zero price was also suggested by a practitioner (and tested in Panel B of their Table 4), indicating that such effects are not merely a theoretical curiosity but arise in real-world implementation contexts.

⁸As an alternative indicator closer in spirit to [Ashraf, Berry, and Shapiro \(2010\)](#), we also express this variable as the transaction price—equal to zero under the full discount and equal to the negotiated price otherwise. Supplementary regressions using this specification are reported in the appendix for robustness.

were therefore never above the prevailing retail price, and the median negotiated price (UGX 7,000) lay well below it. Knowledge of the retail price thus does not mechanically cap the prices we elicit: no farmer was asked to pay more than the seed would cost in the market. Familiarity with the variety and its price, moreover, does not imply certainty about quality: counterfeit and adulterated inputs are widespread in Ugandan seed markets (Bold et al., 2017), so quality uncertainty attaches to the individual bag rather than to the variety, leaving scope for the offer price to serve as a signal about the quality of the specific bag on offer.

Our study sample constitutes 10 farmers per village from 76 villages in Uganda. The study is embedded in a larger factorial design comprising 148 villages in total. Our initial design called for 20 farmers per village across these villages. However, during piloting, we found that information about the bargaining opportunity spread quickly within villages, prompting us to reduce to 10 farmers per village. To compensate, we increased the total number of villages from 110 to 148, of which 76 were assigned to the seed treatment arm reported here (the remaining villages received alternative interventions not analyzed in this paper). Given an intra-cluster correlation of 0.10–0.15 for our primary outcomes (estimated from prior data on 3,450 smallholder maize farmers across 345 villages) this reallocation preserved effective statistical power, as in cluster-randomized designs, additional villages contribute independent variation, while additional farmers within the same village provide diminishing information due to within-village correlation. Simulation-based power calculations in the pre-analysis plan confirm comparable power before and after this design change.

Assignment to surprise discounts was randomized at the village level, as we wanted to avoid that a farmer receiving a bag of seed for free would have close interactions with farmers paying a positive price for the same trial pack. However, we did vary the initial offer price in the bargaining experiment at the individual level. Because the discount assignment varies at the village level, we cluster standard errors at the village level in all regressions.

3 Estimation

To separate the three different effects, we estimate regression models that are similar to the original two-stage design used in for instance Ashraf, Berry, and Shapiro (2010). Recall that in two-stage designs, study participants are given the opportunity to buy a commodity at different price points and in a second stage are given a surprise discount (leading to a proxy for the price the farmer is willing to pay and transaction price at which the farmer obtained the seed after the rebate). In such designs, the outcome variable of interest (for example, an indicator for whether the seed was used as intended) is regressed on both the negotiated price and the transaction price. A statistically significant and positive coefficient on the negotiated price, controlling for the transaction price, provides evidence of a screening effect, since it reflects the positive relationship between the participant’s valuation and the outcome of interest, irrespective of

whether the full price was paid or not. A significant and positive coefficient on the transaction price, controlling for the negotiated price, indicates a sunk cost effect, where paying a higher price increases commitment to using the product, irrespective of the farmer’s valuation of the product.

In our main analysis, we depart from the canonical specification in two ways. First, we add a signaling effect, identified from the randomized initial offer price, which may convey information about unobservable seed attributes such as the quality of the seed. Second, we identify the sunk cost effect using a discontinuous treatment indicator that equals one if the farmer paid the bargained price and zero if the farmer received the full surprise discount. This choice stays closest to the experimental design, where sunk costs were introduced through a randomized, sharp contrast between “free” and “paid” packs. By relying on this exogenous discontinuity, we avoid attributing sunk cost effects to within-paid-group variation in transaction prices that was jointly determined by bargaining and may therefore be endogenous. The interpretation is also straightforward: it captures whether paying anything at all changes subsequent use relative to receiving the pack for free.⁹

In our analysis, we thus estimate the following equation:

$$Y_i = \alpha + \beta_I I_i + \beta_P P_i + \beta_D T_i + \varepsilon_i \quad (1)$$

where Y_i is the outcome of interest (e.g., use of seed from trial pack, adoption of promoted variety in subsequent season,...), I_i is the initial offer price, P_i is the negotiated price (our proxy for willingness to pay), and T_i is a dummy variable indicating if the full price was paid. A statistically significant positive β_I coefficient in equation 1 provides evidence of a signaling effect ($\beta_I > 0$ in equation 1). Evidence of a screening effect is provided by a statistically significant and positive coefficient on WTP ($\beta_P > 0$ in equation 1). Finally, a statistically significant positive coefficient on the dummy variable indicating if the full price was paid provides evidence of a sunk cost effect ($\beta_D > 0$ in equation 1).¹⁰ Because the surprise discount was randomized at the village level, we cluster standard errors at the village level throughout.

One concern with this strategy is that, because bargaining begins from the randomly assigned offer price, that price may anchor the elicited WTP, so that the WTP measure used to identify screening would partly reflect the random

⁹A broader caveat applies to multistage pricing designs. In such designs, the “free” comparison group consists of participants who were first exposed to a price and only subsequently received a surprise discount, rather than participants who received the product without any prior price interaction. Because all participants pass through an initial pricing stage, price exposure itself may influence perceptions and subsequent use. Estimates of sunk cost effects in two-stage designs should therefore be interpreted as the effect of payment conditional on price exposure, rather than as the total effect of paying relative to pure free provision.

¹⁰Note that, in principle, these three tests can be derived from estimating a single equation (Equation 1). However, in the empirical section below we estimate three separate equations, each focusing on the specific effect of interest, and include the other independent variables demeaned and fully interacted with the variable of interest, following the regression adjustment method of [Lin \(2013\)](#), which improves precision and guards against covariate imbalance in randomized experiments.

anchor rather than farmers’ true valuations. In our data, the offer does anchor stated valuations: a one-shilling increase in the initial offer raises elicited WTP by 0.178 shillings ($p < 0.001$). The effect is economically small, however, shifting WTP by only about 7.6% of its mean across the full offer range. Because the offer enters every outcome regression as a control, the WTP coefficient that identifies screening is estimated net of the anchor and is essentially unchanged whether or not the offer is included. Because the offer was randomized independently of the discount, it is orthogonal to the sunk-cost assignment ($p = 0.47$) and does not interact with it ($p = 0.35$), so it cannot confound the sunk-cost estimate. Finally, the offer moves stated valuations but not behavior, with no detectable effect on trial-pack use or subsequent-season adoption: the anchor operates on elicited prices, not on technology use. More details are reported in Online Appendix A.

Our design, based on a fully randomized 100% discount, isolates the sunk cost effect at the extensive margin by comparing farmers who pay their reservation price with those who receive the seed for free. This maximizes statistical power and allows a direct test of the behavioral discontinuity at zero price. By contrast, designs that assign fixed discounts across multiple tiers estimate sunk cost effects along a continuous price gradient, which can be easier to interpret alongside the screening effect—also measured at the margin—and additionally provide insight into how sunk cost effects evolve with payment size rather than only at the free-versus-paid threshold. For completeness, in Appendix B we also present estimates based on the continuous transaction price, which stays closer to the canonical [Ashraf, Berry, and Shapiro \(2010\)](#) two-stage framework. In particular, we estimate Equation 1 again, but instead of T_i being a dummy variable, we now include the actual transaction price, which is zero if the discount was obtained, and equal to the negotiated price if no discount was obtained. Note though that models using the actual transaction price in the context of an all-or-nothing discount are, after controlling for WTP and the randomized offer, effectively reparameterizations with the continuous coefficient mostly rescaling the paid-vs-free contrast by the conditional mean paid price.

In the analysis, we look at different (potentially correlated) outcomes, leading to the problem of multiple hypothesis testing. To deal with this problem, we follow the inverse-covariance weighted (ICW) index method proposed by [Anderson \(2008\)](#) and aggregate different outcome measures within broadly defined families into single summary indices. Each index is computed as a weighted mean of the standardized values of the outcome variables, with the weights derived from the inverse variance covariance matrix of the components of the index. Because the components are standardized to have zero mean and unit variance, a regression coefficient on the index can be interpreted as the effect in units of weighted standard deviations of the component outcomes.

4 Sample and Descriptive Statistics

The sample includes 759 maize-growing households that completed the bargaining experiment, selected to be representative of farmers across four districts in Eastern Uganda: Mayuge, Kamuli, Iganga, and Bugiri. These districts were selected because maize serves as an important crop for both food and income. In these four districts, 76 villages were randomly chosen from a complete list of villages, with the probability of selection proportional to village population size. From each selected village, 10 households were then randomly sampled.

Baseline data collection and the implementation of the experiment took place at the same time in February and March 2023, shortly before the start of the first agricultural season of 2023. The baseline survey asked questions about general household characteristics and more specific questions about farming and seed use. The data was collected by trained enumerators with at least three years of field experience, using standardized protocols that included an introduction and informed consent. Prior to the enumerators' visit for the bargaining experiment, farmers were informed that a team would visit their village and that there might be an opportunity to purchase a product, so they were encouraged to have a small amount of cash available.

Table 1 presents descriptive statistics and balance test results for our Uganda maize sample. The first column reports overall sample means; columns two and three show separate means for the non-discounted and discounted groups; and the Difference column tests whether the village-level discount randomization produced comparable groups. Household heads are on average nearly 50 years old, and about 52 percent have completed primary education. Average household size exceeds eight members. Farmers live, on average, roughly 4 kilometers from the nearest agro-dealer, indicating a relatively dense input supply network.

The two groups are well balanced on most covariates. Household size and the share of male-headed households show statistically significant differences, which is not unexpected given the number of covariates tested. A joint test that regresses the discount assignment on all covariates in Table 1 does not reject orthogonality ($F(11,693)=1.33$, $p=0.21$). Balance also holds for the second, individual-level randomization of the initial offer price: none of the covariates in Table 1 predicts the offer price at the 5 percent level, and the corresponding joint test does not reject orthogonality either ($F(11,693)=1.51$, $p=0.13$). As the surprise discount identifies the sunk-cost effect, we verify that the sunk-cost estimates are robust to controlling for the two imbalanced covariates, as well as to wild cluster bootstrap and randomization inference (Appendix F).

Attrition is minimal. Of the 759 households in the bargaining sample, 98.0 percent were re-interviewed at midline and 98.3 percent at endline. Response rates do not differ by discount assignment, initial offer price, or negotiated price (all p -values above 0.18), and joint tests interacting treatment status with baseline covariates show no evidence of differential attrition. The variation in the number of observations across outcomes in the tables that follow therefore reflects item non-response rather than attrition; one household has a missing negotiated price, so figures based on the bargaining outcome use 758 households.

Table 1: Descriptive Statistics and Balance

	Overall	Not discounted	Discounted	Difference	N
Age household head (years)	48.759 (13.53)	48.024 (14.067)	49.495 (12.949)	1.47 (1.089)	744
Head finished primary education	0.519 (0.5)	0.534 (0.499)	0.504 (0.501)	-0.03 (0.042)	759
Male household head	0.784 (0.412)	0.818 (0.386)	0.749 (0.434)	-0.069* (0.034)	759
Household size	8.167 (3.818)	8.537 (4.169)	7.797 (3.395)	-0.74* (0.342)	759
Distance to agro-dealer (km)	3.812 (4.014)	3.734 (4.46)	3.891 (3.505)	0.158 (0.586)	729
Perceives seed quality as important	0.415 (0.493)	0.432 (0.496)	0.398 (0.49)	-0.033 (0.036)	759
Uses promoted variety	0.063 (0.244)	0.068 (0.253)	0.058 (0.234)	-0.01 (0.02)	759
Seed from formal source	0.35 (0.477)	0.347 (0.477)	0.354 (0.479)	0.006 (0.046)	759
Buys seed often	0.195 (0.396)	0.216 (0.412)	0.174 (0.38)	-0.042 (0.03)	759
Plot size (acres)	1.11 (0.865)	1.141 (0.916)	1.078 (0.812)	-0.063 (0.078)	758
Yield (kg/acre)	421.851 (359.59)	407.313 (364.235)	436.349 (354.789)	29.036 (29.932)	749

Notes: Column 1 reports the overall sample mean with standard deviation in parentheses. Columns 2 and 3 report means with standard deviations in parentheses for the non-discounted and discounted groups. The Difference column reports the OLS coefficient on the discount indicator with village-clustered standard error in parentheses. Rows 1–5 are household characteristics; rows 6–11 are agricultural variables referring to the season prior to the baseline. ⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$.

While the first five characteristics are expected to be unaffected by our intervention, we also report six baseline variables that are more likely to change as a result of the study. To begin, we include a broad measure of technology adoption by asking whether farmers had used seed of any improved maize variety (hybrid seed or open pollinating variety) on any plot during the season prior to the baseline. About 41 percent reported using improved maize varieties.

To capture uptake of the specific varieties promoted in our study, we took a more targeted approach. Farmers were asked to list all plots on which the relevant crop was grown, after which one plot was randomly selected for detailed follow-up questions.¹¹ Only around 6 percent of farmers had already planted the promoted maize variety, Bazooka, in the preceding season. This underscores that the technology introduced in the study had not yet gained traction at scale.

About 35 percent of farmers used maize seed from a formal seed source (for instance, from an agro-input dealer or the government extension system as opposed to own saved or shared between farmers) in the season preceding the baseline. To maintain sufficient vigor, seed, especially hybrid seed, should not be recycled too often. Fewer than 20 percent of maize farmers used seed that had been recycled fewer than five times, which is generally considered the upper limit for open-pollinated varieties. Finally, average maize yields on the randomly selected plot in the season preceding the intervention were around 400 kilograms per acre.

A midline survey was conducted in July 2023, after the first growing season, to capture use of the seed trial pack, agronomic practices, satisfaction with the trial variety, yields, and planning intentions for the next season. The second growing seasons started around August 2023 and harvesting took place in December or January 2024. An endline survey followed in February 2024 to measure actual adoption decisions, complementary input use, and production outcomes in the subsequent season.

5 Analysis

5.1 Price elasticity of demand

A first step in the analysis is to assess whether demand for the seed is responsive to price, since some degree of price sensitivity is a prerequisite for the screening channel. From a policy perspective, the magnitude of the price elasticity of demand indicates how sensitive farmers' purchasing decisions are to price changes and, therefore, how higher prices might constrain initial seed uptake independently of any behavioral mechanisms that influence subsequent use and adoption. In settings where farmers face cash or credit constraints, high price

¹¹The decision to only ask detailed questions on one (randomly selected) plot was guided by the fact that outcomes at plot level (such as adoption of improved inputs and technologies and production outcomes) are likely to be correlated within farm households such that gains in statistical power from surveying all plots likely do not outweigh costs of longer and more tedious questionnaires. As the plot to be surveyed was selected randomly, outcomes should be unbiased and consistent.

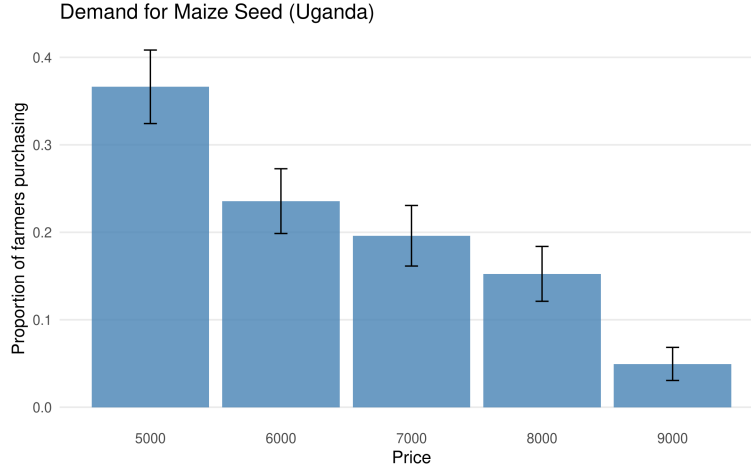


Figure 1: Distribution of negotiated prices. The figure shows the share of small-holder maize farmers (N=758) in Uganda who agreed to purchase the seed trial pack at each final negotiated price during the bargaining experiment. Prices are final agreed prices after bargaining, in Uganda Shillings (UGX).

elasticity may disproportionately exclude those who could potentially benefit most from the technology. In that sense, high price elasticity is often cited as a key justification for offering subsidies. It is important to note that price elasticity captures how many farmers purchase the seed, whereas screening captures how those who purchase differ in their subsequent use.

Figure 1 shows the distribution of prices at which farmers agreed to purchase the maize seed trial pack during the bargaining experiment. We observe a negative relationship between price and the share of farmers who agree to purchase the seed, consistent with a downward-sloping demand curve. Uptake declines steadily as the price increases, indicating relatively high price sensitivity. Using the empirical demand curve, we estimate a point elasticity of demand at the median negotiated price of UGX 7,000 of -2.79 (95 percent confidence interval -3.28 to -2.37, based on a cluster bootstrap over villages): a 1 percent increase in price reduces the share of farmers willing to buy by about 2.8 percent. This pattern confirms that farmers respond to price variation in a way that is consistent with economic theory, and that demand for seed is indeed price-sensitive.

5.2 Three-stage pricing analysis

Having established that demand is price-sensitive, we now test whether the three pricing mechanisms—signaling, screening, and sunk cost—affect how farmers use the seed from the trial pack, whether they plan to use it in the future, and whether they adopt the promoted technology in subsequent seasons. We use regressions to estimate Equation 1 and isolate signaling, screening, and sunk

cost effects of prices.

5.2.1 Trial pack use

We start by testing if, and how, farmers used the seed trial pack. Indeed, the primary outcome in studies that look at the relative importance of screening and sunk cost effects is whether the product that was subsidized or provided for free was used for its intended purpose and not wasted, diverted or sold. In the case of seed trial packs, however, use as intended goes beyond simply planting the seed. The purpose of a trial pack is to enable farmers to learn about a new variety; to observe its performance under local conditions and make an informed decision about whether to adopt it in subsequent seasons. Appropriate use therefore requires that farmers manage the trial in a way that facilitates such learning. This may also involve the use of complementary inputs and recommended agronomic practices, which help ensure that the trial provides an informative signal about the variety’s performance.

Our first family of outcome variables—summarized in a Trial Seed Use Index following [Anderson \(2008\)](#)—is thus comprised of seven outcomes, all measured at midline, immediately after the season when farmers could try out the seed they were offered to buy. First, we simply check if the seed trial pack was used (that is, farmers reported they planted the seed on one for their fields). A second indicator of appropriate use is whether farmers kept the seed separate during planting. Planting the seed separately (rather than mixing it with local varieties) reflects a deliberate intention to evaluate the promoted variety on its own merits, which may influence adoption decisions in subsequent seasons. Similar to the case of planting, we also examine whether farmers kept the harvest from the seed trial pack separate. Doing so indicates that the variety’s identity was maintained throughout the season, enabling an accurate assessment of its yield, market value, and other characteristics such as taste and appearance, and thus serves as an additional measure of appropriate use.

Another indicator of appropriate use concerns who actually planted the seed trial pack. Farmers who pay for the seed are likely to be more motivated and personally invested and may perceive the seed as higher quality, making them more inclined to plant it themselves rather than delegate the task to others. By contrast, those who receive it for free may feel less committed or confident in its value and thus be more willing to pass it on to a family member or hired laborer. Finally, we also measure whether farmers planted the seed correctly—using the recommended spacing and seed number per hole—as a direct indicator of careful engagement with the trial pack. In addition, we include two measures of complementary input use on the plot on which the trial pack was planted: whether the farmer applied fertilizer (organic or inorganic) and whether they used crop protection chemicals (pesticides, insecticides, or fungicides), as complementary inputs are important for the yield benefits of seed of improved varieties to materialize ([Miehe et al., 2025](#)), and their use reflects a deliberate investment in the trial.

Table 2 shows the results. Basic trial pack use is near universal in this

Table 2: Effects on Use of Trial Seed

	sample mean	signaling	screening	sunk cost	nobs
Trial Seed Use Index	0.076 (0.334)	-0.003 (0.011)	0.017** (0.006)	-0.103** (0.037)	669
Used trial pack as seed (1=yes)	0.972 (0.165)	-0.004 (0.004)	0.005 (0.003)	-0.014 (0.014)	749
Kept seed separate (1=yes)	0.94 (0.238)	-0.005 (0.007)	-0.002 (0.004)	-0.003 (0.023)	727
Kept harvest separate (1=yes)	0.713 (0.452)	0.007 (0.016)	0.004 (0.009)	-0.122** (0.047)	683
Buyer used seed personally (1=yes)	0.918 (0.275)	-0.011 (0.009)	0.002 (0.005)	-0.044+ (0.026)	728
Correct planting (1=yes)	0.169 (0.375)	-0.011 (0.012)	0.016* (0.006)	-0.035 (0.036)	714
Used fertilizer (1=yes)	0.546 (0.498)	0 (0.014)	0.021* (0.01)	-0.078 (0.055)	728
Used pesticide (1=yes)	0.285 (0.452)	-0.009 (0.013)	0.01 (0.008)	-0.032 (0.046)	728

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels.

sample: over 97 percent of farmers planted the seed, 94 percent kept the seed separate during planting, and almost 92 percent planted the seed themselves. This confirms that seed trial packs succeed at encouraging experimentation regardless of price.

Comparing treatment groups using the Trial Seed Use Index allows us to distinguish the three potential price mechanisms. We find no evidence of signaling effects. We do find evidence of screening: farmers with higher willingness to pay used the trial pack more appropriately, particularly by following recommended planting practices and applying fertilizer. In contrast, we find a negative sunk cost effect: farmers who paid the full price (i.e., who did not receive the surprise discount) were less likely to use the trial pack in ways that facilitate learning, primarily because they were less likely to keep the harvest separate for evaluation. These estimates are robust to adding baseline wealth and liquidity controls (Appendix E), and inference on the sunk-cost effect is unchanged under a wild cluster bootstrap, randomization inference, and controls for the two imbalanced baseline covariates (Appendix F); the effect on keeping the harvest separate also survives a Romano-Wolf familywise correction across the outcomes in this family.

Appendix Table B.1 presents an alternative specification in which the sunk cost effect is estimated using the transaction price—equal to the farmer’s negotiated price or zero if a full discount was received—instead of the binary indicator distinguishing payers from non-payers. The overall patterns are very

Table 3: Effects on Intentions

	sample mean	signaling	screening	sunk cost	nobs
Adoption Intentions Index	0.282 (0.54)	-0.04 ⁺ (0.021)	0.007 (0.012)	-0.036 (0.051)	473
Plans to use improved variety (1=yes)	0.627 (0.484)	0.009 (0.014)	0.025** (0.009)	0.04 (0.043)	758
Plans to use Bazooka (1=yes)	0.581 (0.494)	0.007 (0.014)	0.023* (0.009)	0.028 (0.047)	758
Planned area for improved variety (acres)	1.514 (1.421)	-0.066 (0.046)	0.019 (0.023)	0.149 (0.156)	473
Plans to buy seed (1=yes)	0.078 (0.268)	-0.011 (0.009)	0.005 (0.005)	-0.032 (0.023)	758

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels.

similar to those reported in the main text, confirming the robustness of our findings: results show a significant screening effect on the Trial Seed Use Index, no significant signaling effects, and a negative sunk cost effect. Appendix Table C.4 reports results after excluding right-censored observations, which yields the same pattern of findings.

5.2.2 Intentions

Beyond actual use of the seed trial pack, we also measured farmers' stated adoption intentions at midline (that is, before the next planting season in which actual adoption is assessed—see next section). These planning outcomes help bridge the gap between how farmers used the trial pack and what they subsequently did in the following season. We examine four outcomes: whether the farmer plans to use seed of an improved crop variety, such as a hybrid or OPV, in the next season; whether the farmer plans to use Bazooka specifically (the seed type used in the experiment); the planned area allocated to seed of improved crop varieties; and whether the farmer plans to purchase fresh seed from an agro-dealer. These measures are aggregated into a Adoption Intentions Index. Table 3 presents the results.

Screening has a positive effect on intentions: Farmers with higher willingness to pay for the seed trial pack are significantly more likely to plan to use seed of an improved maize variety in the next season and are also significantly more likely to plan to use the promoted Bazooka variety specifically. This pattern is consistent with the screening mechanism selecting farmers who are more motivated and better positioned to benefit from the technology. These farmers not only use the trial pack more carefully but also form stronger intentions to continue using improved varieties. We do not find evidence of sunk cost or signaling effects on intended use. Overall, these stated intentions mirror the

screening patterns observed in use behavior, suggesting a behavioral pathway through which screening shapes learning, and intentions. Appendix Table B.2 confirms the robustness of these results using the continuous transaction price specification; Appendix Table C.5 shows similar results after discarding right-censored observations.

5.2.3 Adoption, input use, and production in subsequent season

Adoption of the promoted technology in the subsequent season is particularly relevant, as a primary aim of seed trial packs is to encourage experimentation that may lead to continued use. Observing behavior in the following season provides an early indication of whether farmers’ trial experiences translate into repeated adoption.

The two adoption measures mirror the intention outcomes from the previous section but capture actual behavior on the randomly selected plot: whether seed of an improved variety was used (hybrid or OPV), and whether the promoted Bazooka variety was planted. We also examine whether farmers applied fertilizer or crop protection chemicals on the randomly selected plot, mirroring the complementary input measures from the trial pack season. Finally, we measure production and productivity on the randomly selected plot, both transformed using the inverse hyperbolic sine (IHS) to accommodate zero values while approximating a logarithmic transformation for larger values. As before, we aggregate outcomes into a Subsequent Season Index following [Anderson \(2008\)](#).

Results (in Table 4) reveal a nuanced pattern. Adoption measures show no significant treatment effects: neither willingness to pay (screening) nor the initial offer price (signaling) is associated with the likelihood of using improved or promoted seed in the following season, and complementary input use on the randomly selected plot does not differ across pricing arms. However, screening effects on production are significant and substantial: Farmers with higher willingness to pay achieved significantly higher production on the randomly selected plot in the subsequent season. The association is economically meaningful: a one-standard-deviation increase in willingness to pay (about 2,135 shillings) is associated with roughly 14% higher production, and moving from the 25th to the 75th percentile of willingness to pay (5,000 to 9,000 shillings) with about 27% more. Sunk cost effects on productivity are also significant but negative, mirroring the pattern observed during the trial season: farmers who paid for the trial pack produced about 20% less per unit of land than those who received it for free. This productivity effect is robust to wild cluster bootstrap and randomization inference, although it does not survive a familywise correction across the outcomes in this family (Appendix F).

The interpretation of this association follows from the nature of the negotiated price: it measures effective willingness to pay, in which valuation and resources are bundled by design. When we add baseline wealth and liquidity proxies to the specification (Appendix E), the screening coefficient on trial-pack management is essentially unchanged, indicating that the negotiated price carries information beyond observable resources. The coefficient on subsequent

production attenuates from 0.060 to 0.037 ($p=0.055$), suggesting that for production, part of the screening effect operates through the resources of high-valuation farmers. For a program that uses elicited willingness to pay as a targeting device, both components are relevant: the instrument selects farmers who are both able and motivated to put the technology to productive use. The sunk-cost estimates are unaffected by these controls.

The combination of null adoption effects and significant production and productivity effects may be explained through seed recycling. Among farmers who grew the promoted Bazooka variety on the randomly selected plot in the subsequent season (69% of the sample), 83% used recycled seed from the harvested grain obtained from the seed trial pack, rather than freshly purchased certified seed (despite the fact that a hybrid variety such as Bazooka should not be reused). Recycling rates did not differ across treatment arms (84% among discounted versus 82% among full-price farmers, $p=0.48$), confirming that the production differences are not driven by differential adoption of the variety. Moreover, the screening effect on production remains significant when the sample is restricted to Bazooka recyclers ($p=0.031$), indicating that the effect operates within the group of farmers who recycle the seed trial pack variety. This recycling pattern is itself policy-relevant. A central rationale for distributing trial packs is to seed durable demand for certified seed, yet only about one in six continuing farmers returned to the market for fresh certified seed, even though hybrid maize loses vigor when its grain is replanted; in our setting the packs largely seeded a recycling habit rather than repeat purchase. This shifts where the payoff to the intervention accrues, from repeat certified-seed sales to the quality of the seed farmers save and recycle, and it cautions that the productivity gains we document may erode over successive recycling cycles.

The most likely channel through which pricing affects the quality of recycled seed is through first-season management practices. For the sunk cost channel specifically, farmers who paid for the trial pack saved significantly less grain as seed for the following season ($p=0.039$) and were less likely to keep the trial harvest separate from other varieties during the trial season (Table 2), both of which would reduce the purity and quality of recycled seed. For the screening channel, higher willingness to pay selected farmers who managed the trial more carefully across multiple dimensions, including correct planting, complementary input use, and post-harvest separation. While we cannot isolate which specific management practice drives the subsequent production effect, the evidence is consistent with the view that pricing shapes first-season seed management in ways that affect the quality of recycled seed available for subsequent planting.

Appendix Table B.3 shows that these patterns remain robust when using transaction prices instead of the binary payment indicator. The direction and significance of the coefficients are consistent across specifications, indicating that results are not driven by the binary distinction between paying and receiving the seed for free. Furthermore, Appendix Table C.6 repeats the analysis for non-censored observations, generating similar findings.

Table 4: Effects on Adoption, Input Use, and Production in Subsequent Season

	sample mean	signaling	screening	sunk cost	nobs
Subsequent Season Index	-0.004 (0.589)	0.001 (0.017)	0.009 (0.009)	-0.067 (0.063)	677
Used improved variety on random plot (1=yes)	0.218 (0.413)	0.012 (0.012)	0.006 (0.008)	0.029 (0.043)	703
Used promoted seed on random plot (1=yes)	0.113 (0.317)	0.003 (0.009)	0.005 (0.006)	0.014 (0.031)	703
Used fertilizer on random plot (1=yes)	0.511 (0.5)	-0.002 (0.015)	-0.002 (0.011)	-0.059 (0.053)	697
Used chemicals on random plot (1=yes)	0.401 (0.491)	0.008 (0.013)	-0.005 (0.009)	-0.032 (0.049)	698
Production (IHS)	6.139 (1.113)	-0.031 (0.033)	0.06** (0.02)	-0.147 (0.111)	686
Productivity (IHS)	6.487 (1.068)	-0.035 (0.033)	0.029 (0.018)	-0.223* (0.095)	678

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels.

6 Conclusion

In this paper, we asked whether trial packs of a new agricultural technology should be distributed free of charge or whether charging a positive price makes such interventions more effective. We addressed this question using a field experiment with smallholder maize farmers in Uganda and a three-stage pricing design that separately identifies the screening, sunk-cost, and signaling effects of price. This allows us to move beyond the average effect of charging for a trial pack and identify the distinct mechanisms through which prices shape farmers' behavior.

Our results show that these mechanisms point in different directions. First, we find no evidence that higher prices signal superior product quality or otherwise strengthen farmers' beliefs about the technology. It should be noted that all experimental prices were at or below the prevailing retail price and were offered through a credible project channel, so this null does not rule out price-quality inference in anonymous market transactions, where concerns about counterfeit inputs are most salient. Second, willingness to pay consistently predicts careful use of the trial pack, investment in complementary inputs, and subsequent production, indicating that prices screen for farmers who are better positioned to benefit from a new technology. Third, sunk-cost effects run counter to conventional expectations: farmers who received the seed free of charge managed the trial more carefully than those who paid for it. Rather than reinforcing commitment, payment appears to have reduced experimentation and learning.

These findings highlight that the effects of price are not monolithic. A single price simultaneously determines who receives a technology, how recipients

engage with it, and how much they learn from the experience. More broadly, our results extend the literature on pricing and technology adoption (eg. [Ashraf, Berry, and Shapiro, 2010](#); [Cohen and Dupas, 2010](#)) by showing that prices simultaneously affect selection into use, learning during experimentation, and subsequent productivity. Focusing only on overall adoption therefore obscures distinct mechanisms that may offset one another. In our setting, the positive screening effect of price is offset by a negative effect on experimentation, while signaling plays little role. This decomposition helps explain why charging for trial packs does not increase subsequent adoption despite identifying farmers who ultimately achieve higher productivity.

The results also have broader implications for the design of technology promotion programs. Our findings speak directly to the emerging literature on self-targeting agricultural subsidies ([Duchoslav and De Weerd, 2022](#); [Banda et al., 2022](#)). These proposals rely on the premise that willingness to pay identifies farmers with the highest expected returns to a technology. Our results provide revealed-preference evidence supporting this premise: farmers with higher willingness to pay not only express stronger demand but also manage the technology more carefully and ultimately achieve higher production. Willingness to pay elicited from farmers bidding with their own money is effective willingness to pay: it bundles valuation with wealth and liquidity. For a self-targeting mechanism this is arguably the relevant object, as it is effective demand that a program can elicit and act upon in the field. Controlling for baseline wealth and liquidity proxies leaves the screening effect on trial management essentially unchanged, while the association with subsequent production attenuates, so the elicited price carries information beyond observable resources for some outcomes but not all. A corollary is that targeting on effective demand favors farmers with the means to act on their valuations; programs with an explicit equity objective may therefore wish to combine price-based screening with other targeting criteria.

At the same time, the negative sunk-cost effects suggest that requiring payment may undermine one of the principal objectives of trial packs: encouraging farmers to experiment, observe, and learn under low-risk conditions. These results therefore point toward a hybrid approach, in which willingness to pay is elicited to identify motivated users while the trial pack itself is distributed at a low or zero transaction price. Such an approach preserves the targeting benefits of pricing without sacrificing the learning benefits of free provision.

Taken together, our findings suggest that prices are not simply a tool for rationing demand or recovering costs. They simultaneously determine who experiments with a technology, how that experimentation unfolds, and who ultimately benefits from it. By separating screening, sunk-cost, and signaling effects, this paper shows that these mechanisms need not reinforce one another. In our setting, the value of price lies primarily in identifying farmers most likely to benefit rather than in stimulating adoption itself. This perspective helps reconcile the mixed evidence from the pricing literature by showing that prices simultaneously affect who adopts, how farmers learn, and how technologies are ultimately used.

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Statements and Declarations

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Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics approval

This research received clearance from Makerere’s School of Social Sciences Research Ethics Committee (MAKSSREC 01.23.627/PR1) as well as from IFPRI IRB (DSGD-23-0108). The research was also registered at the Ugandan National Commission for Science and Technology (SS1657ES).

Data and code availability

The data and code used in this study are available at [the project GitHub repository](#).

Use of generative AI in manuscript preparation

During the preparation of this work the authors used ChatGPT (OpenAI, GPT-4 family, accessed via <https://chat.openai.com> between 2025 and 2026) to refine phrasing, improve clarity and structure, and prepare materials for publication and dissemination. The authors also used Claude Code (Anthropic, Claude Opus 4.6 and 4.7, accessed via the Claude Code CLI between 2025 and 2026) to assist with manuscript and reference formatting in Latex. After using these

tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Online Appendix

A Appendix: Anchoring of the Initial Offer Price

This appendix reports the estimates underlying the anchoring discussion in the estimation section. Regressing elicited willingness to pay on the randomized initial offer yields a coefficient of 0.178 ($p < 0.001$), confirming that the offer anchors stated valuations, though by only about 7.6% of mean willingness to pay across the offer range. The initial offer is orthogonal to the randomized discount: regressing the offer on the sunk-cost (paid-versus-free) indicator gives a coefficient of -11.9 shillings ($p = 0.47$), and the offer does not interact with the discount in the sunk-cost regressions ($p = 0.35$). Finally, the screening coefficient is insensitive to the anchor: for the trial-seed use index, the willingness-to-pay coefficient is 0.017 when the initial offer is included as a control and 0.017 when it is omitted.

B Appendix: Estimates Based on the Continuous Transaction Price

Table B.1: Effects on Use of Trial Seed

	sample mean	signaling	screening	sunk cost	nobs
Trial Seed Use Index	0.076 (0.334)	0.001 (0.011)	0.025** (0.006)	-0.014* (0.006)	669
Used trial pack as seed (1=yes)	0.972 (0.165)	-0.003 (0.004)	0.006+ (0.003)	-0.002 (0.002)	749
Kept seed separate (1=yes)	0.94 (0.238)	-0.005 (0.007)	-0.002 (0.005)	0.001 (0.003)	727
Kept harvest separate (1=yes)	0.713 (0.452)	0.007 (0.016)	0.013 (0.009)	-0.018* (0.007)	683
Buyer used seed personally (1=yes)	0.918 (0.275)	-0.01 (0.01)	0.005 (0.005)	-0.006 (0.004)	728
Correct planting (1=yes)	0.169 (0.375)	-0.013 (0.012)	0.019** (0.006)	-0.007 (0.006)	714
Used fertilizer (1=yes)	0.546 (0.498)	0.01 (0.014)	0.027** (0.01)	-0.007 (0.009)	728
Used pesticide (1=yes)	0.285 (0.452)	-0.005 (0.013)	0.013 (0.009)	-0.002 (0.007)	728

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels.

Table B.2: Effects on Intentions

	sample mean	signaling	screening	sunk cost	nobs
Adoption Intentions Index	0.282 (0.54)	-0.042* (0.02)	0.01 (0.012)	-0.005 (0.009)	473
Plans to use improved variety (1=yes)	0.627 (0.484)	0.008 (0.014)	0.022* (0.01)	0.007 (0.007)	758
Plans to use Bazooka (1=yes)	0.581 (0.494)	0.008 (0.014)	0.02* (0.01)	0.005 (0.007)	758
Planned area for improved variety (acres)	1.514 (1.421)	-0.076 (0.052)	0.006 (0.025)	0.031 (0.026)	473
Plans to buy seed (1=yes)	0.078 (0.268)	-0.012 (0.009)	0.007 (0.005)	-0.006 (0.004)	758

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels.

Table B.3: Effects on Adoption, Input Use, and Production in Subsequent Season

	sample mean	signaling	screening	sunk cost	nobs
Subsequent Season Index	-0.004 (0.589)	0.012 (0.019)	0.015 (0.01)	-0.008 (0.01)	677
Used improved variety on random plot (1=yes)	0.218 (0.413)	0.011 (0.013)	0.004 (0.008)	0.005 (0.007)	703
Used promoted seed on random plot (1=yes)	0.113 (0.317)	0.007 (0.009)	0.004 (0.006)	0.002 (0.005)	703
Used fertilizer on random plot (1=yes)	0.511 (0.5)	0.003 (0.017)	0.002 (0.012)	-0.006 (0.008)	697
Used chemicals on random plot (1=yes)	0.401 (0.491)	0.015 (0.014)	-0.002 (0.01)	-0.003 (0.008)	698
Production (IHS)	6.139 (1.113)	-0.017 (0.036)	0.074** (0.023)	-0.022 (0.019)	686
Productivity (IHS)	6.487 (1.068)	-0.024 (0.035)	0.048* (0.02)	-0.036* (0.016)	678

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels.

C Appendix: Excluding Right-Censored Observations

As a robustness check, we re-estimate all main specifications after excluding farmers whose willingness to pay was right-censored---that is, the 118 farmers whose negotiated price equals the randomly assigned initial offer (111 accepted the initial offer outright, while 7 returned to it after bargaining). For these farmers, the observed transaction price represents a lower bound on their true willingness to pay, potentially attenuating the screening estimates. Tables C.4--C.6 report the results. The main findings are qualitatively unchanged. Screening effects on trial seed use remain significant, and the pattern of null adoption and input use results combined with significant production effects in the subsequent season is confirmed. Some individual coefficients lose precision, as expected given the roughly 16 percent smaller sample, but point estimates are stable throughout.

Table C.4: Effects on Use of Trial Seed (Excluding Right-Censored)

	sample mean	signaling	screening	sunk cost	nobs
Trial Seed Use Index	0.076 (0.334)	-0.009 (0.012)	0.023** (0.008)	-0.094* (0.043)	561
Used trial pack as seed (1=yes)	0.972 (0.165)	-0.004 (0.005)	0.008+ (0.004)	-0.018 (0.017)	631
Kept seed separate (1=yes)	0.94 (0.238)	0 (0.009)	-0.005 (0.006)	-0.001 (0.024)	612
Kept harvest separate (1=yes)	0.713 (0.452)	0.003 (0.018)	0.01 (0.011)	-0.125* (0.049)	574
Buyer used seed personally (1=yes)	0.918 (0.275)	-0.016 (0.01)	0.006 (0.006)	-0.031 (0.03)	613
Correct planting (1=yes)	0.169 (0.375)	-0.004 (0.013)	0.014+ (0.008)	-0.059 (0.039)	600
Used fertilizer (1=yes)	0.546 (0.498)	-0.009 (0.016)	0.027+ (0.014)	-0.061 (0.06)	613
Used pesticide (1=yes)	0.285 (0.452)	-0.026 (0.016)	0.023* (0.01)	-0.001 (0.051)	613

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels. Sample excludes 118 farmers whose willingness to pay was right-censored (negotiated price equal to the randomly assigned initial offer).

Table C.5: Effects on Intentions (Excluding Right-Censored)

	sample mean	signaling	screening	sunk cost	nobs
Adoption Intentions Index	0.282 (0.54)	-0.055* (0.025)	0.017 (0.014)	0.007 (0.063)	391
Plans to use improved variety (1=yes)	0.627 (0.484)	0.018 (0.016)	0.025* (0.012)	0.051 (0.047)	639
Plans to use Bazooka (1=yes)	0.581 (0.494)	0.015 (0.016)	0.021+ (0.012)	0.052 (0.051)	639
Planned area for improved variety (acres)	1.514 (1.421)	-0.128+ (0.076)	0.072+ (0.037)	0.253 (0.214)	391
Plans to buy seed (1=yes)	0.078 (0.268)	-0.009 (0.009)	0.004 (0.006)	-0.024 (0.024)	639

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels. Sample excludes 118 farmers whose willingness to pay was right-censored (negotiated price equal to the randomly assigned initial offer).

Table C.6: Effects on Adoption, Input Use, and Production in Subsequent Season (Excluding Right-Censored)

	sample mean	signaling	screening	sunk cost	nobs
Subsequent Season Index	-0.004 (0.589)	-0.021 (0.02)	0.027+ (0.014)	-0.051 (0.07)	572
Used improved variety on random plot (1=yes)	0.218 (0.413)	0.001 (0.013)	0.015 (0.011)	0.038 (0.048)	592
Used promoted seed on random plot (1=yes)	0.113 (0.317)	-0.01 (0.011)	0.012 (0.008)	0.005 (0.033)	592
Used fertilizer on random plot (1=yes)	0.511 (0.5)	-0.004 (0.018)	0.005 (0.014)	-0.037 (0.056)	587
Used chemicals on random plot (1=yes)	0.401 (0.491)	-0.006 (0.018)	0.007 (0.011)	-0.017 (0.058)	588
Production (IHS)	6.139 (1.113)	-0.049 (0.042)	0.079** (0.03)	-0.123 (0.118)	581
Productivity (IHS)	6.487 (1.068)	-0.047 (0.041)	0.031 (0.026)	-0.232* (0.103)	573

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels. Sample excludes 118 farmers whose willingness to pay was right-censored (negotiated price equal to the randomly assigned initial offer).

D Appendix: Balance of the Initial Offer Price Randomization

Table 1 in the main text reports balance for the village-level discount randomization. This appendix reports the corresponding checks for the second randomization: the initial offer price presented at the start of the bargaining game, which was randomized at the individual level. Table D.7 reports, for each baseline covariate in Table 1, the coefficient from a regression of the covariate on the randomly assigned offer price. None of the covariates is related to the offer price at the 5 percent level. A joint test that regresses the offer price on all covariates does not reject orthogonality ($F(11,693)=1.51$, $p=0.13$); the corresponding joint test for the discount randomization is also insignificant ($F(11,693)=1.33$, $p=0.21$).

Table D.7: Balance of the Initial Offer Price Randomization

	coefficient	(SE)	p-value	N
Age household head (years)	0.6797727	(0.413257)	0.1004108	744
Head finished primary education	0.0051628	(0.0167451)	0.7579275	759
Male household head	-0.022571 ⁺	(0.012736)	0.0767594	759
Household size	-0.0381645	(0.1231345)	0.7566908	759
Distance to agro-dealer (km)	-0.1328665	(0.0822791)	0.1067824	729
Perceives seed quality as important	-0.0111936	(0.0150958)	0.458616	759
Uses promoted variety	-0.0113895	(0.0070651)	0.1073612	759
Seed from formal source	0.005251	(0.0143204)	0.7139604	759
Buys seed often	-0.0048339	(0.009877)	0.6246948	759
Plot size (acres)	0.0077505	(0.0269875)	0.7740463	758
Yield (kg/acre)	-0.8611639	(10.8173015)	0.9365689	749

Note: Each row reports the coefficient from a regression of the baseline covariate on the randomly assigned initial offer price (in UGX 1,000), with village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels.

E Appendix: Robustness to Baseline Wealth and Liquidity Controls

Because farmers bid with their own money, the negotiated price measures effective willingness to pay: valuation bundled with wealth, liquidity, and possibly bargaining ability. The uncontrolled estimates in the main text are therefore our preferred estimates of the screening effect. The specifications in this appendix, which add a vector of baseline wealth and liquidity proxies, serve a complementary purpose: they indicate how much of the screening effect operates through observable resources and how much through information in the negotiated price beyond those observables. Table E.8 compares the estimates with and without these controls for the three family indices and the individual outcomes emphasized in the main text. The screening effect on the Trial Seed Use Index is essentially unchanged, as are all sunk-cost coefficients. The screen-

ing coefficients on adoption intentions and on subsequent production attenuate but retain their sign and order of magnitude, suggesting that part of the association between willingness to pay and later outcomes reflects the resources of high-valuation farmers. We interpret this pattern in the main text.

Table E.8: Robustness to Baseline Wealth and Liquidity Controls

	signaling		screening		sunk cost		nobs
	base	+controls	base	+controls	base	+controls	
Trial Seed Use Index	-0.003 (0.011)	-0.005 (0.011)	0.017** (0.006)	0.014* (0.006)	-0.103** (0.037)	-0.104** (0.037)	669
Kept harvest separate (1=yes)	0.007 (0.016)	0.01 (0.016)	0.004 (0.009)	0.007 (0.009)	-0.122** (0.047)	-0.116* (0.048)	683
Plans to use improved variety (1=yes)	0.009 (0.014)	0.006 (0.014)	0.025** (0.009)	0.017+ (0.009)	0.04 (0.043)	0.039 (0.042)	758
Plans to use Bazooka (1=yes)	0.007 (0.013)	0.005 (0.013)	0.023* (0.009)	0.015+ (0.009)	0.028 (0.047)	0.03 (0.046)	758
Adoption Intentions Index	-0.04+ (0.021)	-0.038+ (0.019)	0.007 (0.012)	-0.012 (0.012)	-0.036 (0.051)	-0.056 (0.048)	473
Production (IHS)	-0.031 (0.033)	-0.039 (0.032)	0.06** (0.02)	0.037+ (0.019)	-0.147 (0.111)	-0.17+ (0.102)	686
Productivity (IHS)	-0.035 (0.033)	-0.038 (0.034)	0.029 (0.018)	0.027 (0.017)	-0.223* (0.095)	-0.225* (0.094)	678
Subsequent Season Index	0.001 (0.017)	-0.005 (0.017)	0.009 (0.009)	-0.007 (0.009)	-0.067 (0.063)	-0.071 (0.058)	677

Note: Each pair of columns reports the coefficient from the main specification (base) and from the same specification adding baseline wealth and liquidity proxies: household size, distance to agro-dealer, plot size, total land, number of rooms, an indicator for holding food reserves, group membership, and a weekly consumption expenditure aggregate; continuous controls are mean-imputed with companion missing-value indicators, so sample sizes are identical across specifications. Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels.

F Appendix: Inference Robustness for the Sunk-Cost Effects

The surprise discount that identifies the sunk-cost effect is randomized at the village level, with 38 discounted and 38 non-discounted villages. With this number of clusters, standard cluster-robust inference can understate uncertainty, and Table 1 shows two covariates (household size and male headship) that differ across discount arms. This appendix therefore re-examines inference on the sunk-cost coefficients using a restricted wild cluster bootstrap, randomization inference that permutes the discount assignment across villages, and specifications that control for the imbalanced covariates. Table F.9 reports the results. The three sunk-cost results that are significant under cluster-robust standard er-

rors (the Trial Seed Use Index, keeping the harvest separate, and productivity in the subsequent season) remain significant under all alternative procedures, with p-values between 0.009 and 0.018. All other sunk-cost coefficients are far from significance under every method. We also compute Romano-Wolf familywise-adjusted p-values across the individual outcomes within each family: the effect on keeping the harvest separate survives at the 10 percent level ($p=0.078$), while the productivity effect, though robust in isolation, does not clear the familywise bar against the other five outcomes in its family ($p=0.111$). For completeness, the same familywise correction applied to the screening margin leaves the effects on plans to use an improved variety ($p=0.030$) and on subsequent production ($p=0.023$) significant.

Table F.9: Inference Robustness for the Sunk-Cost Coefficients

	coef.	cluster p	WCB p	RI p	WCB p (ctrl.)	RW p
Trial Seed Use Index	-0.103	0.006	0.013	0.012	0.017	–
Kept harvest separate (1=yes)	-0.122	0.009	0.009	0.012	0.016	0.078
Adoption Intentions Index	-0.036	0.484	0.493	0.507	0.494	–
Production (IHS)	-0.147	0.189	0.217	0.179	0.152	0.631
Productivity (IHS)	-0.223	0.02	0.016	0.018	0.015	0.111
Subsequent Season Index	-0.067	0.293	0.294	0.29	0.251	–

Note: Sunk-cost (paid versus free) coefficients from the main specifications. Columns report p-values from village-clustered standard errors (cluster p), a restricted wild cluster bootstrap with Rademacher weights at the village level (WCB, 999 replications), randomization inference permuting the discount assignment across the 76 villages (RI, 2,000 permutations), the wild cluster bootstrap adding the two baseline covariates that are imbalanced across discount arms as controls (household size and male household head), and Romano-Wolf familywise-adjusted p-values computed across the individual outcomes within each family (RW; not defined for the summary indices).